CHAPTER

05

Informal Fallacies

An argument is good when it meets the criteria of a good argument; when it violates them, it becomes **fallacious**. This chapter explains what a fallacy is, separates **formal** from **informal** fallacies, and then studies the module's **22 informal fallacies** in **five groups**: relevance, weak induction, presumption, ambiguity, and grammatical analogy. Identifying fallacies is the largest single block of marks in the final exam.

LESSONS IN THIS CHAPTER

5.1 Fallacy in General

5.2 Fallacies of Relevance

5.3 Fallacies of Weak Induction

5.4 Fallacies of Presumption

5.5 Fallacies of Ambiguity and Grammatical Analogy

Freshman A+ · Logic & Critical Thinking (PHIL 1011)

CHAPTER 5 · LESSON 5.1

Fallacy in General

AFTER THIS LESSON YOU CAN

- Define a fallacy
- Link fallacies to the criteria of a good argument
- Distinguish formal from informal fallacies
- Name the five groups of informal fallacies

5.1.1 What is a fallacy?

D FALLACY

A defect in an argument that consists in something other than merely false premises. It usually involves **a mistake in reasoning** or **the creation of an illusion that makes a bad argument appear good** (or both).

The module links fallacies to the **four general criteria of a good argument**. A good argument has premises that are:

1. **Relevant** to the truth of the conclusion;
2. **Acceptable** to a rational person;
3. Together **sufficient** for the truth of the conclusion; and
4. Able to give an **effective rebuttal** to all reasonable challenges.

A fallacy is the violation of one or more of these criteria.

★ FALLACIES AND EVALUATION TERMS

Both deductive and inductive arguments can contain fallacies. If they do, they are **unsound** (deductive) or **uncogent** (inductive). Conversely, an unsound or uncogent argument has **a false premise, a fallacy, or both**.

5.1.2 Formal vs informal fallacies

	Formal fallacy	Informal fallacy
Cause	A defect in the **structure (form)** of the argument	A defect in the **content** of the argument (meaning of words, how statements are made, how inferences are drawn)
How detected	By **mere inspection of the form**	Only by **analysing the content**, because informal fallacies hide their true form
Occurs in	**Only deductive** arguments with identifiable forms (categorical, hypothetical, disjunctive syllogisms)	**Both** deductive and inductive arguments

FORMAL FALLACY: CATEGORICAL SYLLOGISM

P1	All tigers are animals. (All A are B)
P2	All mammals are animals. (All C are B)
C	All tigers are mammals. (All A are C)

FORMAL FALLACY: AFFIRMING THE CONSEQUENT

P1	If apes are intelligent, then apes can solve puzzles. (If A, then B)
P2	Apes can solve puzzles. (B)
C	Therefore, apes are intelligent. (A)

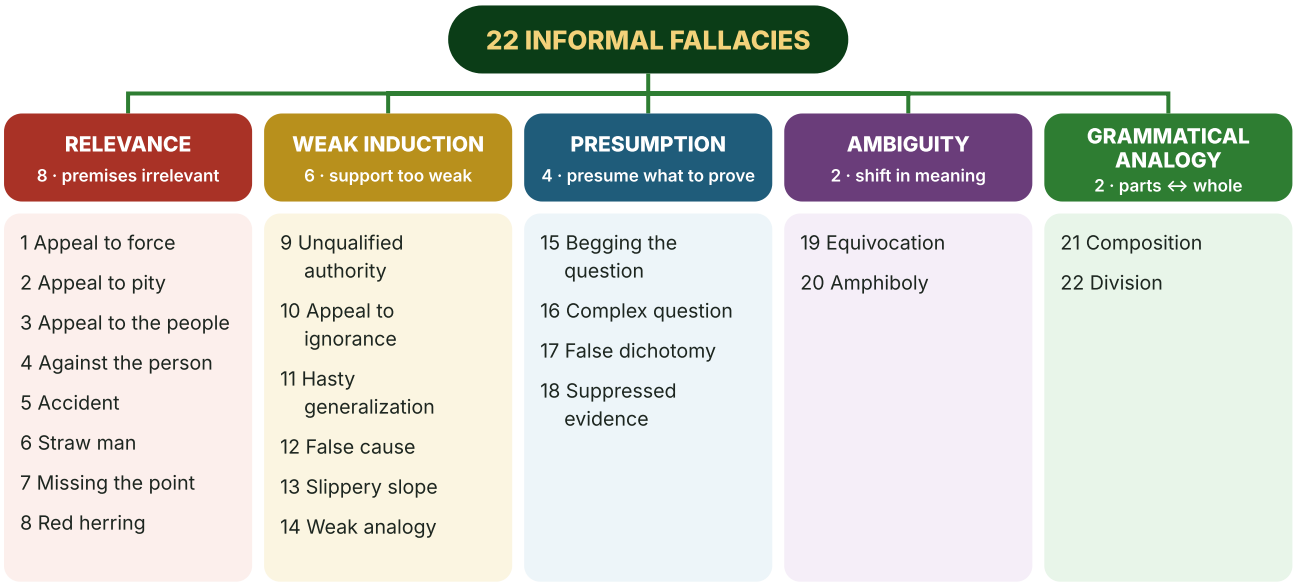
E **AN INFORMAL FALLACY THAT LOOKS VALID**

"All factories are plants. All plants are things that contain chlorophyll. Therefore, all factories are things that contain chlorophyll." It seems to have the valid form "All A are B; all B are C; so all A are C". But "plants" means a **building** in premise 1 and a **life form** in premise 2, so the real form is "All A are B; all C are D; so all A are D", which is **invalid**. Only **content analysis** reveals this.

! **QUICK RULE**

If an argument is **inductive**, it **cannot** contain a formal fallacy. Formal fallacies need a standard deductive form improperly arranged. Breaking a **rule of validity** therefore commits a formal fallacy.

5.1.3 The 22 informal fallacies in five groups



The module's classification: 8 + 6 + 4 + 2 + 2 = 22

M **"8-6-4-2-2": "REALLY WEAK PEOPLE AVOID GRAMMAR"**

Relevance (8) · **W**eak induction (6) · **P**resumption (4) · **A**mbiguity (2) · **G**rammatical analogy (2). Aristotle identified **13** fallacies in two groups; later logicians added dozens more.

Exam Practice

9 QUESTIONS · LESSON 5.1

1 Oda Bultum University · Exam 2

Which one of the following statements is true about fallacy?

A) An argument in which a fallacy occurs is either unsound or uncogent
B) A fallacy is a shortcoming in an argument
C) A formal fallacy occurs only in invalid deductive arguments
D) All

2 Oda Bultum University · Exam 2

Informal fallacies can occur in both deductive and inductive arguments.

True / False

3 Oda Bultum University · Exam 2

A formal fallacy is a kind of fallacy that can be detected only through content analysis.

True / False

4 Oda Bultum University · Exam 2

It is possible to have a fallacy in the absence of an argument.

True / False

5 Salale University · Final 2014

A formal fallacy is committed due to a defect in the content of an argument, not in its structure or form.

True / False

6 Salale University · Final 2015

If an argument violates any one of the rules of validity, it necessarily commits a fallacy.

True / False

7 Salale University · Mid 2015

"If evolution is true, then humans and apes share a common ancestor. They do share one. Therefore, evolution is true." This argument:

A) is valid (modus ponens)
B) is valid (modus tollens)
C) commits affirming the consequent; it is invalid
D) commits denying the antecedent

8 Debre Markos University · Model Exam

Which statement correctly distinguishes formal from informal fallacies?

A) Formal fallacies are detected by inspecting the form; informal fallacies need analysis of content

B) Informal fallacies can always be detected from the form alone

C) There is no difference

D) Formal fallacies occur in inductive arguments; informal fallacies only in deductive ones

9 Debre Markos University · Model Exam

"All factories are plants. All plants contain chlorophyll. So all factories contain chlorophyll." Why is this an informal rather than a formal fallacy?

A) Because its apparent form is invalid

B) Because the word "plants" shifts meaning, and only content analysis reveals it

C) Because it is an inductive argument

D) Because its premises are true

Answers & Explanations

1. D

All

The module supports all three statements.

2. True

Formal fallacies occur only in deductive arguments; informal fallacies can affect both.

3. False

Formal fallacies are detected by inspecting the **form**. Informal fallacies need content analysis.

4. False

A fallacy is a defect **in an argument**.

5. False

That describes an **informal** fallacy. A formal fallacy is a defect in the structure or form.

6. True

Breaking a rule of validity makes the argument invalid because of its form, which is a formal fallacy (e.g., affirming the consequent).

7. C

commits affirming the consequent; it is invalid

Its form is "If A then B; B; therefore A", the formal fallacy the module calls affirming the consequent.

8. A

Formal fallacies are detected by inspecting the form; informal fallacies need analysis of content

Formal fallacies are structural and occur only in deductive arguments. Informal fallacies hide their true form, so only content analysis reveals them.

9. B

Because the word "plants" shifts meaning, and only content analysis reveals it

It looks like the valid form "All A are B; all B are C", but "plants" means a building in one premise and a life form in the other.

CHAPTER 5 · LESSON 5.2

Fallacies of Relevance



AFTER THIS LESSON YOU CAN

- Explain what all fallacies of relevance share
- Identify the eight fallacies of relevance
- Tell legitimate appeals from fallacious ones
- Distinguish straw man, red herring and missing the point

★ **WHAT THEY ALL SHARE**

The **premises are logically irrelevant** to the conclusion but **psychologically relevant**, so the conclusion *seems* to follow. The connection is **emotional**, not logical. Such arguments are called **non sequiturs** ("it does not follow") or **argumentative leaps**. The exception is **missing the point**, where the problem is an **irrelevant conclusion**.

5.2.1 Appeal to force (argumentum ad baculum, "appeal to the stick")

Committed when an arguer tells the listener, explicitly or implicitly, that **some harm will come to him if he does not accept the conclusion**. It always involves a **threat** to the listener's **physical or psychological** well-being.

E **EXAMPLES**

- "Mr. Kebede, drop the fraud charge against me. Remember I am your ex-boss; I will hurt you and your family if you don't."
- "Lately there has been a lot of criticism of our dental benefits policy. If you want to keep working here, you need to know that our policy is fair and reasonable."
  Form: *You can avoid harm by accepting the conclusion → so the conclusion is true.*

! **CAUTION**

The fact that an argument **mentions** a threat does **not** automatically make it a fallacy. The fallacy works by stopping the listener from noticing a missing premise that is false or questionable.

5.2.2 Appeal to pity (argumentum ad misericordiam)

Committed when a conclusion is supported **merely by evoking pity**, and the pitiful facts are **logically unrelated** to the conclusion.

E **FALLACIOUS**

"The headship of the accounting department should be given to Mr. Oumer Abdulla. Oumer has six hungry children to feed and his wife desperately needs an operation to save her eyesight." **Qualification**, not pity, is the relevant reason.

NOT EVERY APPEAL TO PITY IS A FALLACY

Compassion can be **legitimate** when people are in trouble **through no fault of their own**: children orphaned by an earthquake, the elderly, people with disabilities. If people are **responsible** for their own situation, appealing to pity is illegitimate.

5.2.3 Appeal to the people (argumentum ad populum)

It plays on everyone's desire to be **loved, esteemed, admired, recognized and accepted** by others.

Approach	How it works	Example
Direct	Speaking to a **large crowd**, the arguer excites emotions to create a **mob mentality** (propagandists, demagogues)	"I look out at you hard-working people and I know we're right and the critics are wrong!"
Indirect: bandwagon	"**Everyone/the majority** does or believes it, so you should too": the desire to **be part of the group**	"The majority of people accept X, so you should accept X too."
Indirect: vanity	Links a product with **someone admired** (a celebrity), so **you too** will be admired	"Wear the shoe worn by famous footballer Frank Lampard!"
Indirect: snobbery	Appeals to the desire to be **superior, one of the select few**, different from the masses	"Gebeta Guder wine is not for everyone. But you are different, aren't you?"

BANDWAGON VS VANITY VS SNOBBERY

Bandwagon = "**everybody** has it" (join the crowd). **Vanity** = "**a star** has it" (be admired like them). **Snobbery** = "**only a few** can have it" (be above the crowd).

LEGITIMATE APPEALS TO POPULAR OPINION

Not always fallacious: people in a **position to know** (people living near the polar region about when penguins give birth), or a principle widely adopted **after deliberation** (the right to life), carry some weight.

5.2.4 Argument against the person (argumentum ad hominem)

Always involves **two arguers**: one presents an argument, and the other attacks **the person** instead of the argument.

Form	How it works	Recognize it by
Ad hominem abusive	**Verbally abuses** the arguer's character	"Don't accept Abebe's argument for animal protection laws; he is a **divorced drunk** who can't even protect his own family."

Form	How it works	Recognize it by
Ad hominem circumstantial	Points to **circumstances** that supposedly **predispose** the arguer to argue that way	"**Of course** Mr. X argues this way; just look at his circumstances." (Haile Selassie only wanted Italy expelled to regain his power.)
Tu quoque ("you too")	Makes the arguer look **hypocritical**: his behaviour conflicts with his conclusion	"How dare you tell me to quit smoking, doctor? **You** smoke yourself!"

★ **WHEN ATTACKING THE PERSON IS NOT A FALLACY**

When an argument **depends on the person's honesty or credibility**: a lawyer may legitimately question a witness's character for truthfulness (he has lied before). Circumstances can also genuinely bias judgment. Hurley adds that personal comments are relevant when judging **promises, testimony and testimonials**, so *"Hussein is a notorious liar; we should not trust his promises"* commits **no fallacy**.

5.2.5 Accident

Committed when a general rule is applied to a specific case it was not intended to cover. The name comes from the **accidental features** of the case that make it an exception.

E **EXAMPLE**

"Freedom of speech is a constitutionally guaranteed right. Therefore, John Q. Radical should not be arrested for his speech that **incited the riot** last week." Inciting a riot is an exception to the rule.

5.2.6 Straw man

Committed when an arguer distorts an opponent's argument to attack it more easily, demolishes the distorted version, and concludes that the real argument has been demolished. It always involves **two sides**.

E **EXAMPLE**

"Mr. Belay argues that ethnic federalism should be replaced by geographical federalism. But he just wants to take the country back to the past regimes. We don't want to go back to the past, so reject his proposal." Belay's real proposal is **misrepresented** as "going back to the Derg and the monarchy".

To spot it: (1) find the original argument; (2) see how it was distorted (exaggerated, or new assumptions added).

5.2.7 Missing the point (ignoratio elenchi, "ignorance of the proof")

Committed when the premises support one conclusion, but a different conclusion, often vaguely related, is drawn. You should be able to say what the **correct** conclusion would be.

E EXAMPLE

Globalization is making economies interconnected and benefits rich nations at the expense of the poor... "The answer is obvious: poor nations should **detach themselves** from the process." The premises support *redirecting or regulating* globalization, not detaching from it.

5.2.8 Red herring

Committed when the arguer diverts attention by changing the subject to a different but sometimes **subtly related** one, then draws a conclusion about the new issue or simply presumes the point is won. The name comes from dragging a smoked fish across a trail to lead hunting dogs astray.

E TWO TECHNIQUES

- **A subtly related subject:** "The editors of *Addis Flower* newspaper accuse our company of being one of the city's worst water polluters. But *Addis Flower* is responsible for much more pollution than we are; they own a paper company that discharges chemicals into the river."
- **A flashy subject** (sex, crime, scandal): "Professor Conway complains of inadequate parking. But did you know he had a love affair last year...?"

H MODULE VS HURLEY ON THE "POLLUTING NEWSPAPER"

Hurley's **answer key** labels his original version (the *Daily Register*) as **tu quoque**, because the reply makes the accusers look hypocritical. The **module** presents the same example as **red herring**. In your exam, follow the **module**: red herring.

5.2.9 Telling the three "off-track" fallacies apart

	Straw man	Red herring	Missing the point
What happens	**Distorts** the opponent's argument and knocks it down	**Changes the subject**	Draws the **wrong conclusion** from the **original** premises
New premises?	Yes (from the distortion)	Yes (from the new subject)	**No** (uses the original premises)
Conclusion relevant to its premises?	Yes	Yes (if there is one)	**No**
Two arguers?	**Always** (at least implicitly)	Often not	Not necessarily

! CATCH-ALL WARNING

Missing the point serves partly as a **catch-all** fallacy. Do **not** choose it if another fallacy clearly fits.

Exam Practice

37 QUESTIONS · LESSON 5.2

1 Jigjiga University · Final

The fallacy in which the arguer tries to convince by a threat of violence is:

A) Appeal to pity
B) Appeal to force (ad baculum)
C) Appeal to authority
D) Straw man

2 Salale University · Final 2015

"I am certain you will want to stop fighting with our defense force. After all, I'm sure you will want to avoid bombardment of your most populated cities." The fallacy is:

A) Begging the question
B) Appeal to pity
C) Missing the point
D) Appeal to force

3 Oda Bultum University · Exam 3

Secretary to boss: "I deserve a raise. After all, you know how friendly I am with your wife, and I'm sure you wouldn't want her to find out what has been going on between you and that client of yours." The fallacy is:

A) Appeal to pity
B) Appeal to force
C) Hasty generalization
D) Appeal to ignorance

4 Jigjiga University · Final

Which fallacy of relevance do students frequently use with their instructors?

A) Appeal to force
B) Appeal to pity
C) Appeal to vanity
D) Appeal to ignorance

5 Salale University · Final 2014

A student asks the instructor to change an F to a B because otherwise he will be dismissed, his parents are poor, and his life will be meaningless. The fallacy is:

A) Appeal to force
B) Appeal to the people
C) Appeal to pity
D) Bandwagon

6 Oda Bultum University · Exam 3

Taxpayer to judge: "I admit I declared 13 children as dependants though I have only 2. But if you find me guilty, my reputation will be ruined, I will lose my job, my poor wife won't get her operation, and my kids will starve. Surely I am not guilty." The fallacy is:

A) Appeal to force
B) Hasty generalization
C) Appeal to pity
D) Appeal to ignorance

7 Salale University · Final 2014

_____ persuades by appealing to the desire to be loved, esteemed and accepted by others.

A) Appeal to force
B) Appeal to the people
C) Appeal to pity
D) Argument against the person

8 Salale University · Final 2015

"You should watch Tewodros Teshome's latest movie right away. Practically everyone in Addis Ababa is talking about it." The fallacy is:

A) Ad populum: vanity
B) Ad populum: snobbery
C) Ad populum: bandwagon
D) No fallacy

9 Salale University · Final 2014

"It is impossible to be an atheist in a country where 98% of the population believes in God. Thus, you too should believe in God." The fallacy is:

A) Bandwagon
B) Snobbery
C) Vanity
D) Appeal to pity

10 Salale University · Final 2014

"Who is going to buy the new V-8 car, the car first used by the famous Ethiopian athlete Derartu Tulu?" The fallacy is:

A) Bandwagon
B) Appeal to vanity
C) Tu quoque
D) Snobbery

11 Jigjiga University · Final

"Jigjiga International is the best hotel in Jigjiga. It is only for distinguished and respected persons, not for ordinary persons. Please come and enjoy." The fallacy is:

A) Appeal to the people (snobbery form)
B) Appeal to pity
C) Missing the point
D) Accident

12 Oda Bultum University · Exam 4

"Mercedes automobile is not for everyone. If you qualify as one of the select few, buy and drive this distinguished classic car." This commits:

A) Vanity
B) Direct appeal to the people
C) Bandwagon
D) Snobbery

13 Oda Bultum University · Exam 2

"The majority do or believe such and such; therefore you should also do or believe it" is the fallacy of appeal to snobbery.

True / False

14 Salale University · Final 2014

In fallacies of relevance, the premises are both psychologically and logically irrelevant to the conclusion.

True / False

15 Jigjiga University · Final

"Your advice that I should stop smoking is not correct, because you yourself have smoked for several years." The fallacy is:

A) Tu quoque
B) Ad hominem circumstantial
C) Ad hominem abusive
D) Oversimplified cause

16 Salale University · Final 2014

"How can you tell me I should exercise to lose weight? All you do is sit behind a desk all day." The fallacy is:

A) You too (tu quoque)
B) Ad hominem abusive
C) Argument against the person (general)
D) Appeal to ignorance

17 Salale University · Final 2015

"Socrates' philosophy should be dismissed, since the court of Athens found him guilty and sentenced him to death for his stubborn dishonesty." The fallacy is:

A) Ad hominem circumstantial
B) Tu quoque
C) Ad hominem abusive
D) No fallacy

18 Jigjiga University · Final

"Sheikh Mohamed Al Amoudi says Midroc does not push other companies out of the market. But he is the sole owner of Midroc and wants to defend its name. Therefore, we should disregard his argument." The fallacy is:

A) Straw man
B) Ad hominem abusive
C) Ad hominem circumstantial
D) Red herring

19 Salale University · Final 2014

"Saddam Hussein has promised to treat the Kurds and Sunnis with respect. However, Hussein is a notorious liar and an outright murderer. Therefore, we should not trust these promises." The fallacy, if any, is:

A) Red herring
B) Straw man
C) Argument against the person
D) No fallacy

20 Jigjiga University · Final

"The Ethiopian constitution grants everybody the right to demonstrate. Therefore, party X's call for a demonstration at Jigjiga University is correct." The fallacy is:

A) Hasty generalization
B) Accident
C) Division
D) Composition

21 Salale University · Final 2014

"Exceeding the speed limit is illegal. Ambulance drivers always exceed it during emergencies. Thus, their action is illegal." The fallacy is:

A) Straw man
B) Hasty generalization
C) Red herring
D) Accident

22 Jigjiga University · Final

"Senator Maria says she is against tightening the immigration laws. Clearly she does not mind the country being overrun by illegal aliens." The fallacy is:

A) Appeal to the people
B) Against the person
C) Red herring
D) Straw man

23 Oda Bultum University · Exam 2

In which fallacy does an arguer distort another person's argument by making it look more extreme than it really is?

A) Missing the point
B) Straw man
C) Red herring
D) Accident

24 Jigjiga University · Final

"Crimes of theft and robbery have been increasing at an alarming rate. Therefore, we have to kill everyone involved in these crimes." The fallacy is:

A) Missing the point
B) Ad hominem
C) Ad populum
D) Appeal to force

25 Salale University · Final 2015

Lecturers complain about dirty toilets; "the easy solution is to dismiss the complaining lecturers." The fallacy is:

A) Begging the question
B) Appeal to ignorance
C) Straw man
D) Missing the point

26 Jigjiga University · Final

Which fallacy is committed by leading the listener or reader off the track by changing the subject?

A) Straw man
B) Missing the point
C) Slippery slope
D) Red herring

27 Salale University · Final 2015

Both red herring and straw man can lead readers or listeners off the track.

A) True
B) False

28 Oda Bultum University · Exam 2

Which one of the following is different from the others?

A) Appeal to force
B) Appeal to the people
C) Appeal to unqualified authority
D) Appeal to pity

29 Oda Bultum University · Exam 2

Identify the fallacy: "Ato Mohammed has just argued that we should replace the public school system with private education. But of course he argues that way. He has no kids, and he does not want to pay any more taxes for public education."

30 Oda Bultum University · Exam 2

Identify the fallacy: "It is ridiculous to hear that man from Liberia complaining about Ethiopia's poverty. Liberia has twice as much poverty as Ethiopia has ever had."

31 Oda Bultum University · Exam 2

Identify the fallacy: "We've all heard the argument that too much television is the reason our students can't read and write. Yet many of today's TV shows are excellent: medical dramas, comedies, talent shows. Today's TV is just great!"

32 Hurley · Exercise 3.2

"Surely you welcome the opportunity to join our protective organization. Think of all the money you will lose from broken windows, overturned trucks, and damaged merchandise in the event of your not joining." The fallacy is:

A) Appeal to pity
B) Appeal to force
C) Slippery slope
D) Red herring

33 Hurley · Exercise 3.2

"Dan Marino argues that NutriSystem is a great weight-loss program. But that's exactly what you would expect, given that Marino owns stock in NutriSystem. Thus, you shouldn't take his arguments seriously." The fallacy is:

A) Ad hominem abusive
B) Ad hominem circumstantial
C) Tu quoque
D) Unqualified authority

34 Hurley · Exercise 3.2

"TV commentator Larry Kudlow argues that government should get off the back of the American businessman. Obviously, Kudlow wants to abolish government altogether. Without government there would be no defense, no courts... Thus, Kudlow's argument is absurd." The fallacy is:

A) Red herring
B) Missing the point
C) Straw man
D) Slippery slope

35 Hurley · Exercise 3.3

"Environmentalists accuse us of blocking the plan to convert Antarctica into a world park. Nothing could be further from the truth. Antarctica is a huge continent teeming with life: millions of penguins, seals, sea birds..." The fallacy is:

A) Straw man
B) Red herring
C) Missing the point
D) Appeal to ignorance

36 Hurley · Exercise 3.3

"Certainly Miss Malone will be a capable and efficient manager. She has a great figure, a gorgeous face, and tremendous poise, and she dresses very fashionably." The fallacy is:

A) Missing the point
B) Appeal to vanity
C) Hasty generalization
D) Red herring

37 Hurley · Exercise 3.3

"Members of the jury, Shirley Gaines testified that the defendant did not offer to commit the crime. But Gaines is a close friend of the defendant, and only a year ago she was convicted of twelve counts of perjury. Therefore, you should discount Gaines's testimony." The fallacy, if any, is:

A) Ad hominem abusive
B) Ad hominem circumstantial
C) Tu quoque
D) No fallacy

Answers & Explanations

1. **B**

Appeal to force (ad baculum)

Appeal to force always involves a threat to the listener's physical or psychological well-being.

2. D

Appeal to force

It is a veiled threat of harm meant to force acceptance of the conclusion.

3. B

Appeal to force

It is a veiled threat (blackmail). The harm is psychological, which still counts as appeal to force.

4. B

Appeal to pity

Students often ask for better grades by describing sad circumstances rather than showing they earned them. (Appeal to ignorance is a fallacy of weak induction, not relevance.)

5. C

Appeal to pity

The hardship is logically irrelevant to the grade earned, and the student is responsible for his situation.

6. C

Appeal to pity

The pitiful consequences are irrelevant to whether he is guilty of tax evasion.

7. B

Appeal to the people

"Nearly everyone wants to be loved, esteemed, admired, valued, recognized, and accepted by others. The appeal to the people strikes these desires."

8. C

Ad populum: bandwagon

It appeals to the wish to follow the crowd ("everyone is talking about it").

9. A

Bandwagon

It urges a belief **because the majority** holds it.

10. B

Appeal to vanity

It links the product to an **admired celebrity**, so that you too will be admired, like the module's Frank Lampard shoe example.

11. A

Appeal to the people (snobbery form)

It plays on the desire to be regarded as superior: "not for everyone, but you are different".

12. D

Snobbery

It appeals to the wish to be one of a superior "select few".

13. False

That is the **bandwagon** fallacy. Snobbery appeals to a superior few, not the majority.

14. False

The premises are **logically** irrelevant but **psychologically** relevant, which is why the conclusion seems to follow.

15. A

Tu quoque

It tries to make the adviser look hypocritical, as in the module's doctor–patient example.

16. A

You too (tu quoque)

Tu quoque is a kind of argument against the person, but A is the more specific and better answer.

17. C

Ad hominem abusive

It attacks his **character** ("stubborn dishonesty") rather than the content of his philosophy.

18. C

Ad hominem circumstantial

His argument is rejected because of his **circumstances** (owning the company), not because of its content.

19. D

No fallacy

Whether to trust a **promise** depends on the promiser's honesty, so attacking his character for truthfulness is legitimate here. Hurley explains that personal comments are relevant to promises, testimony and testimonials. (Some instructors may still mark C.)

20. B

Accident

A general rule is applied to a specific case it was not intended to cover. The right has limits of place, permission and purpose.

21. D

Accident

Emergencies are an exception the general rule was never meant to cover.

22. D

Straw man

Her real position is distorted into an extreme one, which is then attacked.

23. B

Straw man

The straw man knocks down a distorted version of the opponent's argument.

24. A

Missing the point

The premises support a reasonable conclusion ("take serious measures against theft"), but an extreme, different conclusion is drawn.

25. D

Missing the point

The premises support "improve the cleaning service", but an absurd, different conclusion is drawn.

26. D

Red herring

A straw man also distracts, but by distorting the argument rather than changing the subject.

27. A

True

"The red herring fallacy can be confused with the straw man fallacy because both have the effect of drawing the reader/listener off the track."

28. C

Appeal to unqualified authority

Unqualified authority is a fallacy of **weak induction**; the other three are fallacies of relevance.

29. **Ad hominem circumstantial**

"Of course he argues that way; look at his circumstances."

30. **Tu quoque (you too)**

The complaint is dismissed by pointing at the speaker's own situation instead of being answered. It is a form of argument against the person.

31. **Red herring**

The subject shifts from "does TV harm reading and writing?" to "are TV shows good?". Hurley's answer key: red herring. (The Oda Bultum key said hasty generalization; Hurley's reading fits the definition better.)

32. B

Appeal to force

It is a veiled threat of damage (a protection racket).

33. B

Ad hominem circumstantial

The argument is dismissed by pointing at the arguer's interest (owning stock), not by refuting it.

34. C

Straw man

"Less regulation of business" is distorted into "abolish government altogether", which is then demolished.

35. B

Red herring

The reply drifts to a different subject (Antarctic wildlife) instead of answering the accusation (Hurley: red herring).

36. A

Missing the point

The premises support a conclusion like "she would be a good model", not "a capable manager" (Hurley: missing the point).

37. D

No fallacy

Testimony depends on the witness's honesty, and past perjury and close friendship are legitimate grounds for doubting it (Hurley: no fallacy).

CHAPTER 5 · LESSON 5.3

Fallacies of Weak Induction

AFTER THIS LESSON YOU CAN

- Explain why these are "weak induction" fallacies
- Identify the six fallacies of weak induction
- Know the exceptions to appeal to ignorance
- Distinguish the three kinds of false cause

★ **WHAT THEY SHARE**

These fallacies violate the **sufficiency principle**. The premises are **not logically irrelevant**: they give **at least a shred of evidence**, but **not nearly enough** to make a reasonable person believe the conclusion. Like relevance fallacies, they often involve **emotional** grounds.

5.3.1 Appeal to unqualified authority (argumentum ad verecundiam)

An argument from authority is a normal **inductive** argument. It becomes fallacious when **the authority's reliability can reasonably be doubted**. An authority may be unreliable because he or she:

- **lacks the expertise**;
- is **biased or prejudiced**;
- has a **motive to lie** or spread misinformation;
- lacks the **ability to perceive or recall**.

E **TYPICAL CASES**

- **Celebrities** in advertising: "The famous artist Worku says Vera Pasta is the most nutritious food." Is he an expert on nutrition?
- **Experts outside their field** (subtle, especially when the fields are related): Prof. Kebede, an **animal scientist**, on the division of labour in complex **societies**.
- **Controversial matters** (religion, morality): the authorities themselves disagree, so appeals to authority are weak.

5.3.2 Appeal to ignorance (argumentum ad ignorantiam)

Committed when the premises say nothing has been proved one way or the other, and the conclusion makes a definite claim.

E **BOTH DIRECTIONS ARE FALLACIOUS**

- "No one has ever proved that Haile Selassie I descended from King David. Therefore he did **not**."
- "No one has ever proved that Haile Selassie I did **not** descend from King David. Therefore he **did**."
  The premises tell us only what unnamed people **failed** to do, not positive evidence.

TWO EXCEPTIONS (NO FALLACY)

1. **Qualified researchers** investigate within their expertise and find no evidence → the fruitless search **is** positive evidence. ("Teams of historians failed to show that Tewodros II did not commit suicide at Maqdela → he probably did.")
2. **Courtroom procedure:** a person is **presumed innocent** until proven guilty. If guilt is not proved beyond reasonable doubt, the defendant is "not guilty" in the **legal** sense.

5.3.3 Hasty generalization (converse accident)

A defective inductive generalization. It occurs when **the sample is likely not representative** of the whole group, because it is **too small** or **not randomly selected**.

- **Random sample:** every member of the population has an **equal chance** of being selected.
- **Size:** the larger a random sample, the more closely it matches the population. **Sampling error** is the difference between the frequency of a feature in the sample and in the population (60% in the poll vs 55% in reality = 5% error).
- **Psychological factors** (how questions are asked, who answers) also bias samples.

EXAMPLE

"Addis Zemen interviewed ten young people; none had read a book in the last two years. Conclusion: young people in the country do not have a reading culture."
Note: a small sample is **not always** atypical, and a large one is **not always** typical.

ACCIDENT VS HASTY GENERALIZATION: OPPOSITE DIRECTIONS

Accident applies a **general rule → specific case**. **Hasty generalization** (converse accident) jumps from a **few specific cases → general rule**.

5.3.4 False cause

Committed when the link between premises and conclusion depends on an imagined causal connection that probably does not exist. The best evidence of causation is a **statistical correlation**, but a correlation can be **coincidence** (military spending and infant deaths "explained" like banana consumption), or both events can be caused by a **third, common factor** (pet owners are healthier because sociable people both buy pets and stay healthy).

Variety	Mistake	Example
Post hoc ergo propter hoc ("after this, therefore because of this")	X **came before** Y, so X caused Y. Lies behind most **superstition**	"Since I came into office, violent crime has decreased, so the longer prison sentences we recommended are working." · A black cat crossed my path, then I failed

Variety	Mistake	Example
Non causa pro causa ("not the cause for the cause")	Takes something that is **not really the cause** as the cause, based on something **other than time order** (including mistaking effect for cause)	"Successful executives earn over $100,000. So the best way to make Ferguson a successful executive is to raise his salary." (Success causes salary, not the reverse)
Oversimplified cause	**Many causes** produce the effect, but **one** is picked out as **the sole cause** (often one that is only a minor factor). Usually self-serving	"First-year students' grades are dropping; time on Facebook has increased; so Facebook is the cause." (Ignores parental involvement, the school system...)

H GAMBLER'S FALLACY (ALSO A FALSE CAUSE)

"Emily has bought over 100 lottery tickets and never won. Therefore, the likelihood increases every week that she will win if she keeps buying." Each draw is **independent**, so past losses do not cause future wins.

5.3.5 Slippery slope

A slippery slope argument is a kind of **negative argument from consequences**: "don't take the first step, because it starts a chain of events that ends in disaster." It becomes a **fallacy** when **there is insufficient evidence** that each event in the chain will cause the next.

E EXAMPLE

A Chinese firm runs a donkey slaughterhouse in Bishoftu → workers start eating donkey meat → their families do → the neighbours and village do → the whole country does → Ethiopian food culture collapses. "Therefore the company should be closed." Every link is weak.

★ NOTE

The slippery slope is a **variety of the false cause fallacy** (an imagined chain of causes). It often rests on a mere emotional conviction that something is bad.

5.3.6 Weak analogy

An **argument from analogy** concludes that because two things are similar in some respects, they are similar in another. It is a **fallacy** when the analogy is **not strong enough** to support the conclusion.

★ TWO-STEP TEST

1. List the attributes **a, b, c...** that the two things **share**.
2. Ask how the attribute **z** in the conclusion **relates** to a, b, c. If there is a **causal or systematic relation**, the argument is **strong**; if not, it is **weak**.

STRONG ANALOGY (NO FALLACY)

Substance alpha caused genetic abnormalities in white mice; mice are similar to humans (mammals, similar metabolism), so it probably causes them in humans. The shared biology is **systematically related** to the effect.

WEAK ANALOGY (FALLACY)

Cancer cells are destroyed; murderers and kidnappers are "cancers on society"; so they should be destroyed by capital punishment. But criminals, unlike cells, **can be educated and rehabilitated**.

Exam Practice

27 QUESTIONS · LESSON 5.3

1 Salale University · Mid 2015

The best example of a fallacious argument from authority:

A) My cardiologist says this heart medication is effective for my condition
B) The WHO says this vaccine is safe
C) My favourite actor says this toothpaste is the best
D) The Academy of Sciences says the climate is warming

2 Salale University · Final 2014

"Next Wednesday the Moon will be covered with dust, because the President of our country said so." The fallacy is:

A) Weak analogy
B) Appeal to ignorance
C) Unqualified authority
D) Hasty generalization

3 Salale University · Final 2015

Kino Kabane, a former leader of a Hutu organization, says the Tutsi are not good Rwandans. "On the basis of his authority, we must conclude the Tutsi are senseless." The fallacy is:

A) Argument against the person
B) Amphiboly
C) Straw man
D) Unqualified authority

4 Oda Bultum University · Mid 2017 E.C.

Evaluate: "The famous economist Sir Isaac Newton stated that the World Bank's structural adjustment program cannot be effectively implemented in Third World nations. In view of Newton, we may conclude that the program is ineffective."

5 Oda Bultum University · Exam 3

"No one has ever been able to prove the existence of life after death. We must therefore conclude that life after death is a myth." The fallacy is:

A) Appeal to ignorance
B) Unqualified authority
C) False cause
D) Weak analogy

6 Salale University · Final 2015

"No one knows the whereabouts of the legendary Garden of Eden, so it does not exist." The fallacy is:

A) Missing the point
B) Suppressed evidence
C) Appeal to ignorance
D) Begging the question

7 Jigjiga University · Final

Appeal to ignorance is committed in all of the following ways EXCEPT:

A) Generalizing from an unrepresentative sample
B) Concluding a claim is true because it has not been proved false
C) Concluding a claim is false because it has not been proved true
D) Rejecting a claim because the person who made it habitually lies
E) A and D

8 Salale University · Final 2014

"I've met two people in Dire Dawa so far, and both were friendly. Thus, all the people I meet in Dire Dawa will be nice to me." The fallacy is:

A) Composition
B) Equivocation
C) Amphiboly
D) Hasty generalization

9 Oda Bultum University · Exam 3

"I knew five members of the youth team of the Ethiopian Orthodox Church, and all of them are rigid in outlook. This implies that all followers of the Ethiopian Orthodox Church are conservative." The fallacy is:

A) Composition
B) Ad hominem abusive
C) Accident
D) Hasty generalization

10 Jigjiga University · Final

"A black dog crossed my path, and later I failed the exam." The fallacy is:

A) Non causa pro causa
B) Post hoc ergo propter hoc
C) Slippery slope
D) Oversimplified cause

11 Salale University · Final 2015

"A few minutes after a witch passed our village, a devastating wind blew the roofs off the houses. So we must warn the witch not to pass by." The fallacy is:

A) Suppressed evidence
B) Appeal to ignorance
C) Ad hominem
D) False cause

12 Oda Bultum University · Exam 3

"I told him not to leave the house before the coffee ceremony was over. But he did, and soon after he was arrested by the police. This misfortune would not have fallen upon him if he had not refused me." The fallacy is:

A) Missing the point
B) Post hoc ergo propter hoc
C) Non causa pro causa
D) Suppressed evidence

13 Jigjiga University · Final

Singling out one cause from many possible causes of an event and treating it as the only cause is:

A) Non causa pro causa
B) Post hoc
C) Oversimplified cause
D) Slippery slope

14 Salale University · Final 2015

"Today we can expect a longer life span than our parents and grandparents. Obviously, we owe our thanks to the millions of dedicated doctors." The fallacy is:

A) False cause
B) Appeal to the people
C) Appeal to ignorance
D) Hasty generalization

15 Salale University · Final 2014

"Animal experimentation reduces respect for life → tolerance of war and murder → society becomes a battlefield → the end of civilization. So ban it now." The fallacy is:

A) Slippery slope
B) Red herring
C) Tu quoque
D) Straw man

16 Oda Bultum University · Exam 4

"If we legalize marijuana, next they will legalize cocaine, heroin and meth! Society will collapse!" Thus, we should not legalize marijuana. The fallacy is:

A) Slippery slope
B) False dilemma
C) Equivocation
D) Begging the question

17 Jigjiga University · Final

Arguing that because two things are similar in some respects they are also similar in another respect, when the similarity does not support it, commits:

A) Accident
B) Hasty generalization
C) Weak analogy
D) Equivocation

18 Salale University · Final 2014

Electricity flowing through a wire is compared to water flowing through a pipe, to conclude that voltage is greater at the bottom of a hill. The fallacy is:

A) Slippery slope
B) Complex question
C) Weak analogy
D) False dichotomy

19 Oda Bultum University · Exam 3

"Bonsa was born in Finfinne and is a genius in mathematics. Bikila was also born in Finfinne in the same year. Therefore, Bikila is probably excellent in mathematics." The fallacy is:

A) Weak analogy
B) Appeal to ignorance
C) Suppressed evidence
D) False cause

20 Salale University · Final 2014

Which group of fallacies comes from a lack of enough support from the premises for the conclusion?

A) Fallacies of weak induction
B) Fallacies of relevance
C) Fallacies of grammatical analogy
D) Fallacies of ambiguity

21 Salale University · Final 2014

Which fallacy is NOT committed because the premises are logically irrelevant to the conclusion?

A) Appeal to force
B) Appeal to the people
C) Appeal to unqualified authority
D) Appeal to pity

22 Oda Bultum University · Exam 4

Identify the fallacy, or write "no fallacy": "The county tax collector issued a press release stating that property tax revenues are higher this year than last. Therefore, we conclude that these revenues are indeed higher this year."

23 Hurley · Exercise 3.3

"The secretaries have asked us to provide lounge areas for their coffee breaks. This request must be refused. If we give them lounges, next they'll ask for spas and swimming pools, then racquetball courts, tennis courts and fitness centres. These will drive us into bankruptcy." The fallacy is:

A) Slippery slope
B) Hasty generalization
C) Straw man
D) Appeal to pity

24 Hurley · Exercise 3.3

"Iranian President Ahmadinejad says there are no homosexuals in Iran. Therefore, given his obvious acquaintance with his own country, we must conclude that no gay people live in Iran." The fallacy is:

A) Appeal to ignorance
B) Unqualified authority
C) Hasty generalization
D) Weak analogy

25 Hurley · Exercise 3.3

"On Monday I drank ten rum and Cokes, and next morning I had a headache. On Wednesday, eight gin and Cokes, and a headache. On Friday, nine bourbon and Cokes, and a headache. Obviously, to prevent further headaches I must give up Coke." The fallacy is:

A) Hasty generalization
B) False cause
C) Weak analogy
D) Begging the question

26 Hurley · Exercise 3.3

"Emily has bought over 100 tickets on the weekly lottery and has never won anything. Therefore, the likelihood increases every week that she will win if she continues to buy tickets." The fallacy is:

A) Appeal to ignorance
B) False cause (gambler's fallacy)
C) Hasty generalization
D) Slippery slope

27 Hurley · Exercise 3.3

"To prevent dangerous weapons from being carried aboard airliners, passengers must pass through a magnetometer and submit to a pat-down. Therefore, to prevent alcohol and drugs from being carried into rock concerts, those entering should submit to similar searches." The fallacy is:

A) Weak analogy
B) Accident
C) Slippery slope
D) No fallacy

Answers & Explanations

1. C

My favourite actor says this toothpaste is the best

An actor lacks expertise in dental care. Celebrity endorsements are the module's typical example. A, B and D cite qualified authorities.

2. C

Unqualified authority

A president is not an expert in astronomy.

3. D

Unqualified authority

The cited authority is plainly **biased** against the group. Bias is one of the module's reasons an authority is unreliable.

4. **Inductive, weak, uncogent: appeal to unqualified authority**

Newton was a physicist, not an economist, so the premise is false and the authority is unqualified.

5. A

Appeal to ignorance

It draws a definite conclusion from a **lack of proof**.

6. C

Appeal to ignorance

It concludes "does not exist" from a lack of knowledge.

7. E

A and D

B and C are the two forms of appeal to ignorance. A is hasty generalization and D is ad hominem.

8. D

Hasty generalization

A conclusion about many people is drawn from a sample of **two**.

9. D

Hasty generalization

A small, unrepresentative sample is used to judge a huge group.

10. B

Post hoc ergo propter hoc

It assumes that because the dog came **before** the failure, it caused it, the reasoning behind most superstition.

11. D

False cause

This is post hoc ergo propter hoc, a variety of false cause.

12. B

Post hoc ergo propter hoc

Leaving early came before the arrest, so it is assumed to be the cause.

13. C

Oversimplified cause

Oversimplified cause selects one of many causes as if it were the sole cause.

14. A

False cause

Longer life has many causes (nutrition, sanitation, clean water, vaccines). Giving all the credit to doctors is an **oversimplified cause**.

15. A

Slippery slope

An unlikely chain reaction ending in disaster is assumed, with weak links at every step.

16. A

Slippery slope

It assumes a chain reaction with no evidence for the links.

17. C

Weak analogy

The shared features have no real link to the feature concluded.

18. C

Weak analogy

Water pressure depends on gravity and height; electrical voltage does not. The similarity cannot support the conclusion.

19. A

Weak analogy

Birthplace and birth year have no causal link to mathematical ability.

20. A

Fallacies of weak induction

In these fallacies "the connection between premises and conclusion is not strong enough to support the conclusion."

21. C

Appeal to unqualified authority

It is a fallacy of **weak induction**: the premises give some evidence, but not enough.

22. No fallacy

The tax collector is a **qualified** authority on tax revenues.

23. A

Slippery slope

An unlikely chain from lounges to bankruptcy (Hurley: slippery slope).

24. B

Unqualified authority

The authority is biased and in no position to know such a fact (Hurley: unqualified authority).

25. B

False cause

The common factor that probably causes the headaches is the alcohol, not the Coke (Hurley: false cause).

26. B

False cause (gambler's fallacy)

Each draw is independent, so past losses do not make future wins more likely (Hurley: false cause, gambler's fallacy).

27. A

Weak analogy

An airliner hijacking threatens hundreds of lives; drugs at a concert are a very different danger (Hurley: weak analogy).

CHAPTER 5 · LESSON 5.4

Fallacies of Presumption

AFTER THIS LESSON YOU CAN

- Explain what fallacies of presumption share
- Identify the three forms of begging the question
- Distinguish a complex question from a leading question
- Tell false dichotomy and suppressed evidence from similar fallacies

★ WHAT THEY SHARE

The premises are neither irrelevant nor merely weak. They **presume what they purport to prove**:
- **Begging the question** presumes the premises are **adequate**.
- **Complex question** presumes a question can be answered by **"yes", "no"** or a brief answer.
- **False dichotomy** presumes an "either ... or" statement is **jointly exhaustive**.
- **Suppressed evidence** presumes **no important evidence** has been overlooked.

5.4.1 Begging the question (petitio principii, "request for the source")

Committed when the arguer creates the illusion that inadequate premises provide adequate support in one of three ways. After hearing it you want to ask: *"But how do you know X?"*

Form	How	Module examples
1. Leaving out a possibly false key premise (most common)	Uses phrases like "**of course**", "**clearly**", "**obviously**", "**this being the case**", "**after all**" to hide the gap	"Murder is morally wrong. This being the case, abortion is morally wrong." (missing: *abortion is murder*) · "The world displays amazing organization. **Obviously** it was created by an intelligent God." (missing: *organization can only come from an intelligent creator*)
2. Restating the premise as the conclusion	The conclusion says the same thing as a premise **in different words**	"Capital punishment is **justified** because it is **legitimate and appropriate** to put such criminals to death." · "Anyone who preaches revolution has a vision of the future, because if a person has no vision of the future he could not possibly preach revolution."
3. Circular reasoning	A chain of inferences that **comes back to its starting point**	"Harar Brewery produces the finest beer because they have the best chemists, because they can pay more, because they produce the finest beer."

! "FAMOUS" = "WELL KNOWN"

"All of us cannot be famous, because all of us cannot be well known." → **begging the question** (form 2). Also: "Butane is combustible. Therefore, it burns." and "Students below a 2.0 GPA should not graduate, because students who fail to score 2.0 cannot graduate."

5.4.2 Complex question

Committed when two (or more) questions are asked as one, and a single answer is given to both. Every complex question **presumes a condition**. Answering it in either direction produces an argument that "establishes" that condition, and the respondent is **trapped**.

E "HAVE YOU STOPPED CHEATING ON EXAMS?"

- "Yes" → *so you cheated in the past.* "No" → *so you are still cheating.*
- It is really two questions: **Did you cheat in the past? If so, have you stopped?** The correct response is to **split the question and answer each part separately**.

! COMPLEX QUESTION VS LEADING QUESTION

A **leading question** suggests its answer ("Tell us, on April 9, did you see the defendant shoot the deceased?") but involves **no logical fallacy**: it does not try to trick the respondent into admitting something. To tell them apart, you may need to know whether earlier questions were asked.

5.4.3 False dichotomy (false bifurcation, the "either–or" fallacy)

Committed when an "either ... or" premise presents two unlikely alternatives as if they were the only ones; the arguer then eliminates the undesirable one, leaving the desired conclusion.

! FALLACY

"Either you buy me a new car or I will divorce you. You don't want a divorce. So you have to buy me a car."
"Classical democracy originated either from the Gada system or from Athens. It did not originate from Athens. So it originated from the Gada system."

★ NO FALLACY

"Either the Abay River is in Ethiopia or it is in South Africa. It is not in South Africa. Therefore, it is in Ethiopia."
If one of the alternatives is actually true, no fallacy is committed.

★ VALID BUT UNSOUND

The form ("Either A or B; not A; therefore B") is **valid**. The problem is that the disjunctive premise is **false** (or probably false), so the argument is **unsound**. The fallacy makes a false premise **appear true**. Usually the arguer states only the either–or premise and leaves the rest to the listener.

5.4.4 Suppressed evidence

A **cogent** argument must have true premises that **do not ignore important evidence** that outweighs the given evidence and supports a different conclusion. **If an inductive argument ignores such evidence, it commits suppressed evidence.**

E **FOUR COMMON FORMS**

1. **Ignoring a key fact:** "Somalia is good for investment: cheap raw materials, cheap labour, a good market, a port..." (ignores the lack of **peace and stability**).
2. **Advertisements:** "Buy a bucket of chicken and have a barrel of fun!" (the fun is not in the bucket, and the fat isn't mentioned).
3. **Ignoring changes over time:** "Poland had a low standard of living for fifty years, so it will for the next fifty" (ignores the collapse of the Soviet bloc).
4. **Quoting out of context:** the US Second Amendment quoted without its "well-regulated militia" clause.

! **SUPPRESSED EVIDENCE VS BEGGING THE QUESTION (FORM 1)**

Both leave out a premise. **Begging the question** omits a premise that is **needed to support** the stated conclusion. **Suppressed evidence** omits a premise that would **require a different conclusion**. They sometimes overlap.

Exam Practice

19 QUESTIONS · LESSON 5.4

1 Jigjiga University · Final

Mr. John said, "You have only two options: either love it or leave it." The fallacy is:

A) Suppressed evidence
B) False dichotomy
C) Begging the question
D) Complex question

2 Salale University · Final 2015

"Whoever is not with us is against us" is an example of a false dichotomy.

True / False

3 Salale University · Final 2014

"Those who are not with us are our enemies. You are not with us. Therefore, you are our enemy." The fallacy is:

A) False cause
B) Post hoc
C) False dichotomy
D) Straw man

4 Salale University · Final 2015

Mercy: "Either you care for and contribute to your children, or you are not a good father. Your wife told me you neither contribute nor care. So you are not a good father." The fallacy, if any:

A) False dichotomy
B) Suppressed evidence
C) Unqualified authority
D) No fallacy

5 Oda Bultum University · Exam 4

"Either we ban all guns, or criminals will take over the country." The fallacy is:

A) False dichotomy
B) Appeal to ignorance
C) Red herring
D) Post hoc

6 Jigjiga University · Final

"All of us cannot be famous, because all of us cannot be well known." The fallacy is:

A) Begging the question
B) Suppressed evidence
C) Complex question
D) Oversimplified cause

7 Salale University · Final 2015

"Students whose GPA is below 2.0 should not graduate, because students who fail to score 2.0 cannot graduate." The fallacy is:

A) Argument against the person
B) Suppressed evidence
C) Begging the question
D) No fallacy

8 Salale University · Final 2014

A fallacy in which the conclusion is stated in one of the premises is called:

A) Appeal to vanity
B) Begging the question
C) Complex question
D) Appeal to ignorance

9 Oda Bultum University · Exam 4

Identify the fallacy: "The world in which we live displays an amazing degree of organization. Obviously this world was created by an intelligent God."

10 Oda Bultum University · Exam 4

Identify the fallacy: "Butane is combustible. Therefore, it burns."

11 Salale University · Final 2014

Which fallacy is intended to trap the respondent into acknowledging something they might not want to acknowledge?

A) False dichotomy
B) Equivocation
C) Amphiboly
D) Complex question

12 Oda Bultum University · Exam 4

"Have you stopped smoking marijuana?" The fallacy is:

A) You too
B) Straw man
C) Slippery slope
D) Complex question

13 Jigjiga University · Final

When the arguer reaches a conclusion by ignoring evidence that outweighs the presented evidence, the fallacy is:

A) Appeal to ignorance
B) False dichotomy
C) Suppressed evidence
D) Complex question

14 Salale University · Final 2015

"In the past 30 years, Ethiopians enjoyed peace, stability and security. Therefore, they will probably enjoy them for the next 30 years." The fallacy is:

A) Hasty generalization
B) Suppressed evidence
C) Slippery slope
D) Composition

15 Salale University · Final 2014

Begging the question is classified under which group of fallacies?

A) Fallacies of relevance
B) Fallacies of weak induction
C) Fallacies of presumption
D) Fallacies of ambiguity

16 Oda Bultum University · Exam 4

Identify the fallacy: "Either Bekele Gerba or Andualem Gosa is a popular singer. Bekele Gerba is not a popular singer. Therefore, Andualem Gosa is a popular singer."

17 Hurley · Exercise 3.4

"Either we require forced sterilization of Third World peoples or the world population will explode and all of us will die. We certainly don't want to die, so we must require forced sterilization." The fallacy is:

A) Slippery slope
B) False dichotomy
C) Appeal to force
D) Suppressed evidence

18 Hurley · Exercise 3.4

"People who lack humility have no sense of beauty, because everyone who has a sense of beauty also has humility." The fallacy is:

A) Begging the question
B) Equivocation
C) Hasty generalization
D) Composition

19 Hurley · Exercise 3.4

"During the fifty years Mr. Jones worked, he contributed $90,000 to Social Security. Now he stands to collect $200,000. Obviously he will collect a much greater monetary value than he contributed." The fallacy is:

A) Suppressed evidence
B) Begging the question
C) False cause
D) Division

Answers & Explanations

1. B

False dichotomy

Two options are presented as if they were the only ones; one could stay and try to change things.

2. True

It ignores other options, such as being neutral or undecided.

3. C

False dichotomy

It assumes only two options (with us, or an enemy).

4. D

No fallacy

It is a valid disjunctive syllogism, and the disjunction is reasonable: one alternative is clearly true. The module says that if one alternative is true, no fallacy is committed. (Some keys say A; this book follows the module's rule.)

5. A

False dichotomy

It presents two extreme options as if they were the only ones.

6. A

Begging the question

"Famous" and "well known" mean the same thing, so the conclusion just restates the premise (form 2).

7. C

Begging the question

The premise restates the conclusion in slightly different words.

8. B

Begging the question

Restating the conclusion as a premise is the second form of begging the question.

9. Begging the question

It leaves out the key premise that organization can only come from an intelligent creator, hiding the gap with "obviously".

10. Begging the question

"Combustible" means "burns", so the conclusion just restates the premise.

11. D

Complex question

For example, "Have you stopped cheating on exams?"

12. D

Complex question

It hides two questions: Did you smoke? Have you stopped?

13. C

Suppressed evidence

This is the module's definition of suppressed evidence.

14. B

Suppressed evidence

It ignores recent events and changing conditions that make the prediction improbable, the same pattern as the module's Poland example.

15. C

Fallacies of presumption

The four fallacies of presumption are begging the question, complex question, false dichotomy and suppressed evidence.

16. False dichotomy

The "either ... or" premise is false, because neither of them is a singer. It makes a false premise look true.

17. B

False dichotomy

Many other options exist, such as education and voluntary family planning (Hurley: false dichotomy).

18. A

Begging the question

The premise says the same thing as the conclusion in different words (Hurley: begging the question).

19. A

Suppressed evidence

It ignores inflation and the interest his contributions would have earned over fifty years (Hurley: suppressed evidence).

CHAPTER 5 · LESSON 5.5

Fallacies of Ambiguity and Grammatical Analogy

AFTER THIS LESSON YOU CAN

- Explain equivocation and amphiboly and how they differ
- Explain composition and division
- Tell composition from hasty generalization and division from accident
- Recognize legitimate part–whole transfers

5.5.1 Fallacies of ambiguity

These arise when **the conclusion depends on a shift in the meaning of an ambiguous word or phrase**, or on **a wrong interpretation of an ambiguous statement**. They violate the **clarity principle**.

19 · Equivocation

Committed when a word or phrase is used in two different senses in the argument. Such arguments are either invalid or have a false premise, so they are **unsound**.

E MODULE EXAMPLES

- "Some triangles are **obtuse**. Whatever is **obtuse** is ignorant. Therefore, some triangles are ignorant." (angle vs dull-minded)
- "Any **law** can be repealed by the legislature. The **law** of gravity is a law. So the law of gravity can be repealed." (statutory law vs law of nature)
- "We have a duty to do what is **right**. We have a **right** to speak out for the innocent. So we have a duty to speak out." (morally correct vs a just claim)

To be convincing, the two senses must be **subtly related** (only the "right" example might fool someone), or the shift must be **spread across a long argument** ("equal opportunity", "national security" in political speeches).

20 · Amphiboly

Committed when the arguer misinterprets an ambiguous statement and draws a conclusion from the faulty interpretation. The statement is usually made by **someone else**, and the ambiguity comes from **grammar or punctuation**: a missing comma, a **dangling modifier**, an **ambiguous pronoun antecedent**. This is **syntactical ambiguity**.

E MODULE EXAMPLES

- "The tour guide said that **standing in Meskel Square**, the new federal police building could easily be seen." (Who is standing there, the observer or the building?)
- "Habtom told Megeressa that **he** had made a mistake. So Habtom has the courage to admit his mistakes." (Who is "he"?)
- Wills and contracts: "I leave my house and my clothes to Lemma and Mengistu" (respectively or collectively?)

	Equivocation	Amphiboly
Source of ambiguity	A **word or phrase** with two meanings	The **structure (syntax)** of a whole statement
Who creates it	Usually **the arguer's own** creation	Usually the arguer **misinterprets someone else's** statement
Both at once?	"The cookbook recommends serving the oysters when thoroughly **stewed**..." → **amphiboly** (oysters or diners?) **and equivocation** (cooked vs drunk)	

5.5.2 Fallacies of grammatical analogy

These arguments are **grammatically similar** (analogous) to other arguments that are good, so they **appear good yet are bad**.

21 · Composition (parts → whole)

Committed when an attribute is wrongly transferred from the parts of something onto the whole.

! FALLACIOUS

- Each player on this team is excellent → the **team** is excellent.
- Each atom in this chalk is invisible → the **chalk** is invisible.
- Sodium and chlorine are deadly poisons → **salt** is a deadly poison.

★ LEGITIMATE (NO FALLACY)

- Every atom in this chalk has **mass** → the chalk has mass.
- Every component of this fence is **white** → the whole fence is white.
- Each cell of this carrot is 90% water → the carrot is 90% water.

22 · Division (whole → parts)

The exact reverse of composition: an attribute is wrongly transferred from a whole (or class) onto its parts (or members).

! FALLACIOUS

- Salt is non-poisonous → **sodium and chlorine** are non-poisonous.
- The Royal Society is over 300 years old; General Merid Hussein is a member → **he** is over 300 years old.
- Water extinguishes fire → **hydrogen and oxygen** extinguish fire.

★ LEGITIMATE (NO FALLACY)

- This chalk has mass → its **atoms** have mass.
- This field of poppies is uniformly orange → the **individual poppies** are orange.

5.5.3 The two hardest distinctions

★ **DISTRIBUTIVE VS COLLECTIVE PREDICATION**

- **"Fleas are small"** → a **general statement**: the attribute applies to **each and every** flea (**distributively**).
- **"Fleas are numerous"** → a **class statement**: the attribute applies to the **class as a whole** (**collectively**), not to each flea.

If the argument...	Look at...	Then it is...
Moves from **members → class**	the **conclusion**	General statement → **hasty generalization** · Class statement → **composition**
Moves from **class → members**	the **premises**	General statement → **accident** · Class statement → **division**

E **CARS AND TRUCKS**

"Less gasoline is consumed by a car than by a truck. Therefore, less gasoline is consumed in the United States by cars than by trucks." The conclusion is a **class statement** (the whole class of cars uses less), so this is **composition**, not hasty generalization. (It is false, because there are many more cars.)

Exam Practice

19 QUESTIONS · LESSON 5.5

1 Salale University · Final 2015

The fallacy of equivocation arises from a syntactical defect in a statement.

True / False

2 Salale University · Final 2015

Amphiboly usually arises from the arguer misinterpreting a statement made by someone else.

True / False

3 Salale University · Final 2014

The fallacy of composition is committed when we improperly transfer the attributes of the whole to its parts.

True / False

4 Jigjiga University · Final

Identify the correct statement about fallacies of ambiguity.

A) The premises seem relevant, but the conclusion depends on a shift in meaning
B) The connection between premises and conclusion is emotional
C) The premises are not strong enough to support the conclusion
D) The premises presume what they purport to prove

5 Jigjiga University · Final

"Abeba told me that he always quarrels with his father when he is drunk. Therefore, Abeba's father is quarrelsome whenever he drinks." The fallacy is:

A) Amphiboly
B) Equivocation
C) Red herring
D) Division

6 Salale University · Final 2014

"Mulugeta told Fikadu that he always quarrels with his father when he is drunk." Which linguistic fallacy is involved?

A) Equivocation
B) Division
C) Composition
D) Amphiboly

7 Salale University · Final 2015

"Kebede said he was interviewing for a job drilling oil wells in the supervisor's office. So the supervisor must have an awfully dirty office." The fallacy is:

A) Amphiboly
B) Straw man
C) Red herring
D) Equivocation

8 Jigjiga University · Final

"Water can extinguish fire. So its components, hydrogen and oxygen, can extinguish fire." The fallacy is:

A) Amphiboly
B) Division
C) Composition
D) Equivocation

9 Salale University · Final 2015

"The students at Bradford College come from every one of the fifty states. Michelle attends Bradford College. Therefore, Michelle comes from every one of the fifty states." The fallacy is:

A) Division
B) Composition
C) Accident
D) No fallacy

10 Oda Bultum University · Exam 3

"Every member of the Delta Club is over 70 years old. Therefore, the Delta Club must be over 70 years old." The fallacy is:

A) Suppressed evidence
B) Amphiboly
C) Begging the question
D) Composition

11 Oda Bultum University · Exam 3

"Ethiopian Airlines is the most efficient and progressive organization in Ethiopia. Bonsa is an employee of Ethiopian Airlines. Therefore, he must be a progressive man too." The fallacy is:

A) Composition
B) Division
C) Hasty generalization
D) Converse accident

12 Oda Bultum University · Exam 4

"This food is very delicious. Therefore, every component in this food is delicious." The fallacy is:

A) Equivocation
B) Composition
C) Division
D) Amphiboly

13 Oda Bultum University · Exam 4

Identify the fallacy, or write "no fallacy": "This field of poppies is uniformly orange. Therefore, the individual poppies are orange."

14 Oda Bultum University · Exam 4

Identify the fallacy, or write "no fallacy": "Each and every cell in this carrot is 90 percent water. Therefore, the entire carrot is 90 percent water."

15 Hurley · Exercise 3.4

"James said that he saw a picture of a beautiful girl stashed in Stephen's locker. We can only conclude that Stephen has broken the rules, because girls are not allowed in the locker room." The fallacy is:

A) Equivocation
B) Amphiboly
C) Composition
D) Begging the question

16 Hurley · Exercise 3.4

"If Thomas gives Marie a ring, then Thomas and Marie will be engaged. Thomas did give Marie a ring. In fact, he phoned her just the other night. Therefore, they are engaged." The fallacy is:

A) Amphiboly
B) Equivocation
C) False cause
D) No fallacy

17 Hurley · Exercise 3.4

"All men are mortal. Therefore, some day man will disappear from the earth." The fallacy is:

A) Division
B) Composition
C) Hasty generalization
D) Accident

18 Hurley · Exercise 3.4

"Switzerland is 48 percent Protestant. Heidi Gilsing is a Swiss. Therefore, Heidi Gilsing is 48 percent Protestant." The fallacy is:

A) Composition
B) Division
C) Accident
D) Equivocation

19 Bule Hora University · Model Exam

"Less gasoline is consumed by a car than by a truck. Therefore, less gasoline is consumed in Ethiopia by cars than by trucks." The fallacy is:

A) No fallacy
B) Composition
C) Hasty generalization
D) Division

Answers & Explanations

1. False

Equivocation comes from a **word or phrase** used in two senses. A syntactical defect causes **amphiboly**.

2. True

"Amphiboly usually involves a mistake made by the arguer in interpreting an ambiguous statement made by someone else."

3. False

That is **division**. Composition goes from the **parts** to the **whole**.

4. A

The premises seem relevant, but the conclusion depends on a shift in meaning

B describes fallacies of relevance, C fallacies of weak induction, and D fallacies of presumption. (The original option A read "the premises are logically relevant to the conclusion"; it is the best match by elimination.)

5. A

Amphiboly

The pronoun "he" has an ambiguous antecedent (Abeba or the father), a syntactical ambiguity like the module's Habtom example.

6. D

Amphiboly

"He" could refer to Mulugeta, to Fikadu, or even to the father.

7. A

Amphiboly

"In the supervisor's office" can attach to "interviewing" or to "drilling". The arguer picks the unintended reading.

8. B

Division

An attribute of the whole is wrongly transferred to its parts (in fact hydrogen burns and oxygen feeds fire).

9. A

Division

The premise is a **class** statement (about the student body as a whole), and its attribute is wrongly given to one member. A general rule would make it accident.

10. D

Composition

An attribute of the parts (members) is wrongly transferred to the whole (the club).

11. B

Division

An attribute of the whole organization is wrongly transferred to one member.

12. C

Division

Whole → parts: think of raw salt or raw spices on their own.

13. No fallacy

This transfer from whole to parts is legitimate: a uniformly orange field is orange because each poppy is.

14. No fallacy

This transfer from parts to whole is legitimate.

15. B

Amphiboly

Was the **picture** or the **girl** in the locker? The arguer chooses the wrong reading of an ambiguous sentence (Hurley: amphiboly).

16. B

Equivocation

"Ring" shifts from a piece of jewellery to a phone call (Hurley: equivocation).

17. B

Composition

Mortality belongs to each individual (distributively); the conclusion transfers it to the human race as a whole (collectively) (Hurley: composition).

18. B

Division

A class statement about Switzerland as a whole is transferred to one member (Hurley: division).

19. B

Composition

The conclusion is a **class** statement (the whole class of cars), so it is composition, not hasty generalization.

CHAPTER 5 · QUICK REVISION

Informal Fallacies — at a glance

The 22 informal fallacies: master table

#	Fallacy	Group	Spot it by...
1	Appeal to force	Relevance	A **threat** of harm
2	Appeal to pity	Relevance	**Sad circumstances** irrelevant to the issue
3	Appeal to the people	Relevance	Crowd emotion (direct), **everyone** (bandwagon), **a star** (vanity), **the select few** (snobbery)
4	Against the person	Relevance	Attacks **character** (abusive), **circumstances** (circumstantial), **hypocrisy** (tu quoque)
5	Accident	Relevance	**General rule → exceptional case**
6	Straw man	Relevance	**Distorts** the opponent's view, then attacks it
7	Missing the point	Relevance	Premises support a **different conclusion**
8	Red herring	Relevance	**Changes the subject**
9	Unqualified authority	Weak induction	Authority not expert, **biased**, or lying
10	Appeal to ignorance	Weak induction	"**Not proved** → so false / so true"
11	Hasty generalization	Weak induction	**Small or biased sample → all**
12	False cause	Weak induction	Post hoc (**before** → cause) · non causa pro causa (not the cause) · oversimplified (**one of many** causes)
13	Slippery slope	Weak induction	An unlikely **chain reaction** to disaster
14	Weak analogy	Weak induction	Similarity **unrelated** to the conclusion
15	Begging the question	Presumption	Missing key premise ("**obviously**") · restating · circle
16	Complex question	Presumption	**Two questions in one** ("Have you stopped...?")
17	False dichotomy	Presumption	"**Either ... or**" with false, non-exhaustive options
18	Suppressed evidence	Presumption	**Ignores** important outweighing evidence
19	Equivocation	Ambiguity	**One word**, two meanings
20	Amphiboly	Ambiguity	**Sentence structure** (grammar, punctuation, pronoun)

#	Fallacy	Group	Spot it by...
21	Composition	Grammatical analogy	**Parts → whole**
22	Division	Grammatical analogy	**Whole → parts**

★ A 4-STEP METHOD FOR ANY "IDENTIFY THE FALLACY" QUESTION

1. **Is there a threat, pity, a crowd, or a personal attack?** → relevance (1–4).
2. **Is a rule misapplied, a view distorted, the subject changed, or the conclusion off target?** → accident, straw man, red herring, missing the point.
3. **Is the support just too weak** (authority, ignorance, sample, cause, chain, analogy)? → weak induction (9–14).
4. **Does it assume what it should prove** (circle, trick question, either–or, hidden evidence)? Or does it turn on **word meaning** or **parts and wholes**? → presumption, ambiguity, grammatical analogy.

Always check for "no fallacy" (qualified authority, legitimate part–whole transfer, a true either–or, a witness's credibility).

! FINAL EXAM TRAPS

- **Accident** (general → specific) vs **hasty generalization** (specific → general).
- **Composition** (parts → whole) vs **division** (whole → parts).
- **Equivocation** (word) vs **amphiboly** (sentence).
- **Bandwagon** (everyone) vs **vanity** (a celebrity) vs **snobbery** (the select few).
- **Abusive** (character) vs **circumstantial** (interests) vs **tu quoque** (hypocrisy).
- **Straw man** (distorts) vs **red herring** (changes the subject) vs **missing the point** (wrong conclusion from the original premises).
- **Post hoc** (time order) vs **non causa pro causa** (not the cause) vs **oversimplified** (one of many).
- Appeal to unqualified authority is **weak induction**, not relevance.